

ODD END SEMESTER EXAMINATION DECEMBER-2025

Roll no.....

Name of the Course: B.Tech, CE (Specialization in RS & GIS)

Semester: V

Name of the Paper: Advance Geographical Information systems

Paper Code: TCE 599

Time: 3 Hour

Maximum Marks: 100

Note:

- (i) All questions are compulsory.
- (ii) Answer any two sub question among a, b & C in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks.

Q1.	(20 Marks). CO1
a. Describe the procedure involved in performing a line intersection analysis within GIS datasets.	
b. Using suitable diagrams, demonstrate how Union and Intersection operations are carried out on polygonal features.	
c. Discuss why conversion between raster and vector data formats is essential for comprehensive spatial analysis.	
Q2.	(20 Marks), CO2
a. Explain the distinction between spatial information and attribute (non-spatial) information using appropriate examples.	
b. Compare and contrast the raster data model and vector data model based on structure, resolution, and storage considerations.	
c. Summarize various methods used for analyzing attribute data within a GIS environment.	
Q3.	(20 Marks). CO3
a. Define the concept of spatial modelling and justify its importance in resource analysis.	
b. Illustrate the applications of GIS in environmental and resource management.	
c. Evaluate the usefulness of AM/FM systems in managing utility networks such as power distribution.	
Q4.	(20 Marks). CO4
a. Explain different approaches to maintain data reliability and quality control in GIS databases.	
b. Discuss the types and implications of spatial resolution in RS-GIS data interpretation.	
c. Propose techniques to improve attribute data precision and reduce uncertainty in GIS projects.	
Q5.	(20 Marks). CO5
a. Construct a multi-criteria analytical approach for groundwater-quality evaluation using GIS datasets.	
b. Design a spatial decision-support system for groundwater potential mapping based on thematic criteria.	
c. Develop a GIS-based methodology for land-suitability assessment for specific agricultural crops.	

